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Bihar Engineering University, Patna

B.Tech 8th Semester Exam., 2022

Code: 105801 ***Time: 03 Hours***

Semester: VIII ***Subject: DATA MINING*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option (any seven) : ***[2 x 7 = 14]***

(a) The artifacts are
- (i) deterministic distortions
- (ii) non-deterministic distortions
- (iii) deterministic and non-deterministic distortions
- (iv) None of the above

(b) Outlier are
- (i) legitimate data objects
- (ii) illegitimate data objects
- (iii) legitimate and illegitimate data objects
- (iv) None of the above

(c) Jaccard coefficient is frequently used to handle objects consisting of
- (i) symmetric binary attribute
- (ii) asymmetric binary attribute
- (iii) Both (i) and (ii)
- (iv) None of the above

(d) The 0 (zero) correlation value between two data object indicates
- (i) no linear relationship between two objects
- (ii) non-linear relationship between two objects
- (iii) linear relationship between two objects
- (iv) Both (i) and (ii)

(e) The decision boundaries produced in nearest-neighbor classifiers are
- (i) rectilinear
- (ii) oblique
- (iii) arbitrary
- (iv) None of the above

(f) The margin of the decision boundary in SVM is given by
- (i) $d = 1/\|w\|$
- (ii) $d = 1/(2 * \|w\|)$
- (iii) $d = 2/\|w\|$
- (iv) $d = 1/\|w\|^2$

(g) The classification and prediction methods are affected by
- (i) accuracy
- (ii) speed
- (iii) robustness
- (iv) All of the above

(h) The total number of possible rule extracted from a data set that contains d item is
- (i) $R = 3^{d+1} - 2^d + 1$
- (ii) $R = 3^{d+1} - 2^{d+1} + 1$
- (iii) $R = 3^d - 2^{d+1} + 1$
- (iv) $R = 3^d - 2^d + 1$

(i) The computational complexity of Apriori algorithm increases with the ____ in the bound of support threshold.
- (i) increase
- (ii) decrease
- (iii) Does not depend
- (iv) None of the above

(j) The training time is ____ and testing time is ____ of ANN.
- (i) low, high
- (ii) high, low
- (iii) low, low
- (iv) high, high

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Q.2 ***[7+7=14]***

(a) What is data mining? Differentiate data mining with traditional database system.

(b) In real-world data, tuples with duplicate and redundant values for some attributes are a common occurrence. Describe various methods for handling this problem.

Q.3 ***[14]***

A database has five transactions. Let *min sup* = 60% and *min conf* = 80% :

TID	items_bought
T100	{M, O, N, K, E, Y}
T200	{D, O, N, K, E, Y}
T300	{M, A, K, E}
T400	{M, U, C, K, Y}
T500	{C, O, O, K, I, E}

Find all frequent item sets using Apriori and FP-growth, respectively. Compare the efficiency of the two mining processes.

Q.4 ***[7+7=14]***

(a) Why is naive Bayesian classification called 'naive'? Briefly outline the major ideas of naive Bayesian classification.

(b) What is boosting? State why it may improve the accuracy of decision tree induction.

Q.5 ***[7+7=14]***

(a) What is data classification? How does it differ from prediction?

(b) Describe the ID3 algorithm for decision tree construction. Why is it unsuitable for decision tree construction?

Q.6 ***[14]***

The support vector machine (SVM) is a highly accurate classification method. However, SVM classifiers suffer from slow processing when training with a large set of data tuples. Discuss how to overcome this difficulty and develop a scalable SVM algorithm for efficient SVM classification in large data sets.

Q.7 ***[14]***

Briefly describe the following approaches to clustering :
partitioning methods, hierarchical methods, density-based methods, grid-based methods, model-based methods, methods for high-dimensional data and constraint-based methods
Give examples in each case.

Q.8 ***[14]***

What are the differences between the three main types of data warehouse usage : information processing, analytical processing and data mining? Discuss the motivation behind OLAP mining (OLAM).

Q.9 ***[7+7=14]***

(a) For the following vectors, x and y, calculate the indicated similarity or distance measures :
(i) $x = (0, -1, 0, 1)$, $y = (1, 0, -1, 0)$ cosine, correlation
(ii) $x = (0, 1, 0, 1)$, $y = (1, 0, 1, 0)$ Euclidean, SMC

(b) Discuss overfitting and underfitting in decision tree construction with suitable example.

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